

Beaconhouse National University

BS Computer Science

Final Year Project

DAILYKHATA

Deliverable 1

Target Market & Audience Analysis

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Academic Year: 2026-2027

Version 2.0 | 23 September 2026

Document Control

Project: DailyKhata

Document: Target Market & Audience Analysis

Version: 2.0

Status: Authoritative reconstruction baseline under the user-authorised scope

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Supervisor: Nouman Ali

Academic year: 2026-2027

Revision date: 23 September 2026

Submission date: Not recorded for this replacement

Supervisor endorsement: To be recorded after review

Revision History

Historical source versions remain unchanged in their original locations. This replacement consolidates scope, terminology and evidence provenance under the recorded decisions. Version 2.0, 23 September 2026, prepared for the DailyKhata Project Team.

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1 Purpose and project foundation

DailyKhata addresses the effort of maintaining useful personal and household expense records. This deliverable establishes its intended audience, user problems, needs, expectations and target-market justification. It is the factual and scope foundation inherited by Deliverable 2. It does not claim an implemented system or completed user validation. [A1, U1]

DailyKhata is a personal expense management application primarily designed for Pakistani users who frequently deal with paper receipts, cash transactions and informal expense tracking. It reduces the effort required to maintain personal or household expense records through receipt digitisation, OCR, quick cash expense capture, manual expense entry, automatic categorisation, user confirmation and correction, a structured Chart of Accounts, transaction history, spending summaries, item-level price history, semantic product classification and an AI Financial Assistant.

2 Market and Pakistani context

The industry is financial technology, focused on personal expense management. The relevant market is expense-management applications for Pakistani users. The selected segment comprises adults who manage personal or household spending, use smartphones, and regularly encounter paper receipts or cash purchases without receipts. The original documents use an indicative age range of 18 to 50; this is a design focus, not a verified market statistic or a maximum user age. [S1, S2]

A grocery receipt can contain several products that later need different expense categories. A rickshaw fare may leave no receipt at all. A notebook or phone note can preserve an amount, but the user still has to maintain consistent entries and assemble totals. DailyKhata combines capture and review with structured records so that the same information supports later search, summaries and questions. This is project-team analysis of the reported workflows, not a claim that every Pakistani household uses the same method.

2.1 Evidence and market claim limits

The strongest available project evidence is the original D2 narrative: three reported interviews, informal observation and a reported comparison of existing applications. It describes effort, delayed entry and distrust of incorrect categories. No interview transcripts, observation counts, recordings, survey data or competitor test artefacts accompany that narrative. These findings support a small qualitative design rationale, not prevalence or market-size estimates. D2 retains the detailed evidence register. [S2, pp. 3, 10-12]

The original estimates of several million potential users and typical monthly spending of Rs. 40,000 are not retained as facts because no supporting calculation or source is present. Statements that all competing applications require manual entry or lack price history are also not adopted. The market opportunity is framed as a testable fit between the selected

users and DailyKhata's workflow, not a verified claim of market uniqueness. [DEC-08, DEC-09]

3 Primary and secondary users

Primary users are homemakers and household budget managers, university students and salaried professionals. Small shopkeepers are secondary direct users for petty cash, supplier expenses, receipt capture and expense summaries. Family members and advisers are secondary beneficiaries of information shared by the account holder.

The baseline is a single-account expense-management system. Household expenses can be recorded by the account holder and reviewed together. CSV and/or PDF export is Should Have; a dedicated exportable household summary is Could Have. Shared household accounts, invitations, household permissions and collaborative editing are not current requirements.

3.1 Current exclusions

Direct bank integration; automatic debit-card or credit-card synchronisation; Easypaisa and JazzCash transaction integration; open banking; payment processing; money transfers; inventory management; point-of-sale functionality; payroll; enterprise ERP; complete SME accounting; tax filing; investment management; credit scoring; fraud-detection infrastructure; commercial banking functionality; enterprise multi-tenant organisation management; banking-grade regulatory certification; production infrastructure for millions of concurrent users; arbitrary third-party financial integrations.

3.2 Future scope

Bank transaction importing; wallet integrations; shared household accounts; collaborative budgeting; business accounting features; inventory; POS; recurring expenses; budgeting goals; predictive analytics; anomaly detection; advanced financial recommendations; deeper Urdu support; additional languages; multi-currency support; retailer integrations; e-receipt ingestion.

A feature can be excluded from the current project and also be named as potential future work. Future listing is not a delivery promise. Recording an expense paid by a method described by the user does not create a bank or wallet integration.

4 Canonical personas

These are canonical design personas, not a new participant sample. Names and source-supported attributes are preserved from D1 and D2. Usman Raza uses the D1 design age of 41; D2 reports an interview age of 46 without corroborating records. DEC-01 records that unresolved source fact without leaving two competing persona definitions.

Table 1: Canonical personas

ID and name	Age and role	Classification
P-01 Ayesha Tariq	34 Homemaker and household budget manager	Primary
P-02 Bilal Ahmed	27 Salaried professional	Primary
P-03 Hamza Sheikh	21 University student	Primary
P-04 Zubair Hassan	Not recorded Family member and spouse of a primary user	Secondary beneficiary
P-05 Usman Raza	41 Small shopkeeper	Secondary direct user

4.1 Ayesha Tariq

Goal: Track weekly household spending and grocery prices. Behaviour: Handles paper receipts and cash; prefers a photograph to transcription. Needs readable review, familiar categories and price history; expects correction and privacy.

4.2 Bilal Ahmed

Goal: Reconcile and understand monthly personal expenses. Behaviour: Comfortable with digital tools; reviews spending periodically. Needs searchable history and plain-language answers; expects minimal upkeep.

4.3 Hamza Sheikh

Goal: Track personal spending and allowance with little effort. Behaviour: Frequent small cash purchases; low tolerance for repeated form entry. Needs quick text entry without a receipt; expects useful category totals.

4.4 Zubair Hassan

Goal: Review household spending with the account holder. Behaviour: Occasional review of summaries shared by the account holder. Needs understandable totals; no separate household login or permissions implied.

4.5 Usman Raza

Goal: Track petty cash and supplier expenses. Behaviour: Uses the same receipt and expense workflow for a general store. Needs dependable expense records; sales, inventory and POS are excluded.

Digital confidence varies within every user group. Persona descriptions guide design and recruitment; they do not justify assuming that all homemakers have low digital literacy or

all students have high technical skill. Bilal and Zubair are not reported interview participants. Zubair's age is not recorded and no age is inferred.

5 User problems needs and expectations

Paper receipts and unrecorded cash purchases leave personal and household expense histories incomplete. Manual upkeep makes consistent recording difficult. Users need low-effort capture and reliable ways to review and understand spending.

Table 2: User problems linked to needs and benefits

Problem and source	Need and response	Expected benefit
Scattered or missing receipt records [S1; S2 p. 3]	Receipt digitisation, reviewed OCR and searchable history	Less transcription and a retrievable structured record
Small no-receipt cash purchases omitted [S2 pp. 10-11]	Manual quick entry and text-based cash entry	Record spending at purchase time or afterwards
Manual classification and incorrect categories [S2 p. 10]	Automatic categorisation with confirmation and correction	Reduce repeated work while retaining user control
Difficulty reconstructing totals [S2 pp. 3, 10]	Category totals, monthly summaries and dashboard	Understand recorded spending without rereading every note
No usable item comparison in reported workflow [S1; S2 p. 10]	Item-level price history and semantic product classification	Compare recorded products without losing brand identity
Effort finding an answer [S1]	AI Financial Assistant grounded in authorised records	Ask plain-language questions about spending
Sensitive financial records [S1; U1]	Authentication, access controls and secure storage	Keep each user's data private

Users expect low effort, understandable values and reliable correction. Automation should reduce transcription rather than hide uncertainty. The record is saved after review, and saved transactions remain editable and deletable. A failed OCR attempt should lead to a clear retry or manual-entry path. Receipt image archiving is optional, so structured record retention must not be described as permanent preservation of every receipt image.

Fast interaction is an expectation, not a measured result. The source interview question about logging in under 10 seconds establishes interest in low-effort capture, not a validated end-to-end benchmark. Later evaluation should separate simple cash entry from image processing and user correction. Privacy, ordinary-device usability and coherent totals are essential to trust. [DEC-10, DEC-11]

6 Authoritative capability baseline

The following register is shared by D1, D2 and Project Scope. Must Have means required for the minimum successful FYP; Should Have means intended but deferrable; Could Have means optional if time permits. Priority is not a claim of implementation.

Table 3: Canonical feature scope and priorities

ID	Capability	Priority
SC-01	User registration and authentication	Must Have
SC-02	Basic user profile and settings	Must Have
SC-03	Personal digital expense ledger	Must Have
SC-04	Photograph or upload receipts	Must Have
SC-05	OCR processing	Must Have
SC-06	Structured extraction of merchant and date where available, line items, quantities where available, prices and receipt total	Must Have
SC-07	Review OCR results and edit incorrect extracted values before saving	Must Have
SC-08	Manual quick expense entry	Must Have
SC-09	Text-based cash expense entry	Must Have
SC-10	Voice-based cash expense entry	Should Have
SC-11	Automatic transaction and line-item categorisation	Must Have
SC-12	User confirmation or correction of AI-generated categories	Must Have
SC-13	Locally understandable expense category labels	Must Have
SC-14	Full internal Chart of Accounts	Must Have
SC-15	Mapping of user-facing categories to accounting accounts	Must Have
SC-16	Accounting-compatible transaction storage	Must Have
SC-17	Transaction history	Must Have
SC-18	Search and filtering	Must Have
SC-19	Category-level spending totals	Must Have
SC-20	Basic monthly spending summaries	Must Have
SC-21	Item-level price history	Must Have
SC-22	Semantic product classification	Must Have
SC-23	Natural-language financial querying	Must Have

ID	Capability	Priority
SC-24	AI Financial Assistant	Must Have
SC-25	Dashboard	Must Have
SC-26	Editing and deleting saved expense transactions	Must Have
SC-27	Secure user-data storage	Must Have
SC-28	English and Urdu localisation	Should Have
SC-29	CSV and/or PDF export	Should Have
SC-30	Reasonable support for ordinary Android-class devices	Must Have
SS-03	Custom category labels	Should Have
SS-04	Simple spending visualisations	Should Have
SS-05	Month-to-month comparison views	Should Have
SS-06	Receipt image archive	Should Have
SS-08	User-created categories	Could Have
SS-09	Offline draft expense entry	Could Have
SS-10	Expense reminders	Could Have
SS-11	Exportable household summaries	Could Have

Legacy aliases are retained for traceability: SS-01 refers to SC-10, SS-02 to SC-29, and SS-07 to SC-28. They do not create additional features. Month-to-month comparison views are optional; answering period comparisons through the AI Financial Assistant is core.

7 Foundational design concepts

7.1 Accounting and category boundaries

DailyKhata must maintain a full internal Chart of Accounts with stable account identifiers, account types and mappings from user-facing expense categories. Full means a coherent internal account framework, including the account classes needed for structured recording, not a complete SME accounting interface. Only expense-management workflows are committed. A local label is separate from its category identity and account mapping.

Table 4: Illustrative local category mappings

User-facing category	Internal accounting account
Ration	Groceries Expense
Transport	Transport Expense
Dining	Dining Expense
Utilities	Utilities Expense
Healthcare	Healthcare Expense
Education	Education Expense
Entertainment	Entertainment Expense
Household Supplies	Household Supplies Expense

These mappings illustrate the approved architecture; account codes and the final hierarchy are later design decisions. For groceries purchased in cash for Rs. 2,500, a double-entry implementation debits Groceries Expense by Rs. 2,500 and credits Cash by Rs. 2,500. Every journal must balance wherever double-entry is used. Users should not enter journals or account codes. No accounting or regulatory certification is claimed.

A mixed receipt retains its receipt total and separately categorised line items. Category totals must not count the same purchase once at transaction level and again at line-item level. Edits and deletion must also update dependent totals and price history consistently. Accounting corrections must preserve balance where journals are used. These are project-team design constraints derived from the need for trustworthy records.

7.2 Semantic product classification

Original receipt text, individual product identity, brand, canonical product type, expense category and accounting classification are separate concepts. Preserve the original text alongside corrected values and inferred classifications. A canonical product type groups related products without replacing their identities.

Table 5: Illustrative semantic product distinctions

Original text and brand	Product identity and type	Category and account
Surf Excel 1kg Brand: Surf Excel	Distinct Surf Excel 1kg product Laundry Detergent	Household Supplies Household Supplies Expense
Bonus 1kg Brand: Bonus	Distinct Bonus 1kg product Laundry Detergent	Household Supplies Household Supplies Expense

Both products can support analysis of Laundry Detergent spending. Exact price history still distinguishes brand, product and pack size. Comparisons require comparable quantities and units; where these are missing, show recorded prices with a limitation

instead of claiming equivalent value. Uncertain product matches remain reviewable. Semantic similarity is not proof that two products are identical.

7.3 AI Financial Assistant

The AI Financial Assistant is a core conversational capability grounded in the current user's authorised DailyKhata records. It supports natural-language questions, spending summaries, comparisons between periods and categories, recurring-spending analysis, product-price comparisons, spending patterns, unusually high recorded spending, explanations of change and possible saving opportunities. It may give personalised general financial suggestions based on available historical data.

Illustrative response, not research: restaurant spending of Rs. 8,400 this month compared with Rs. 5,900 last month gives a difference of Rs. 2,500. The assistant may explain that returning to the earlier level would reduce spending by that amount. It must distinguish recorded facts, calculations and suggestions, state the relevant period, and acknowledge missing data. It does not move money, execute payments or control accounts.

Conversational analysis of repeated past expenses is core; automatically generating recurring expenses is future scope. Data-grounded observations about high spending are core; predictive anomaly detection and advanced financial recommendation systems remain future scope. The team may integrate existing models and services instead of training a foundation model.

7.4 Localisation and delivery limits

English and Urdu localisation is Should Have. The architecture should permit translated labels, right-to-left Urdu presentation, local terminology, Urdu-friendly interaction and potential Urdu interaction with the AI Financial Assistant. An initial English interface does not prove bilingual localisation. Urdu support is not guaranteed until implemented and validated. Local category labels remain Must Have independently of translation.

The application is mobile-first and should support ordinary Android-class devices. Receipt OCR targets reasonably clear printed receipts in supported formats, with correction and manual fallback for failures. No universal receipt accuracy or response-time threshold has been validated. Frameworks, model providers, device test specifications and measurable performance targets are later decisions. Offline draft entry is Could Have and does not imply offline OCR or offline assistant services.

8 Target market justification and feasibility

The selected groups face related recording tasks at different scales. Ayesha needs weekly household visibility, Hamza needs a short path for frequent cash purchases, and Bilal needs a coherent monthly view. One expense ledger, supported by receipt and text capture, can serve these needs without separate enterprise workflows. Usman can reuse that same ledger for expenses without changing the primary market. [S1, S2; project-team analysis]

A four-student team can divide work into capture and review, ledger and accounting mappings, product semantics and price history, and assistant and presentation, while sharing integration and testing. These are proposed work areas, not assignments to named students. Using existing OCR and language-model components is permitted. The academic contribution lies in integration, structured data, correction, semantic modelling, traceability and evaluation. Optional features should follow a working end-to-end Must Have workflow.

The fit should be tested with representative users and receipts. Useful validation includes whether users can correct extraction, record a no-receipt expense, find a past transaction and understand an assistant answer. Record actual task completion and error causes before setting quantitative acceptance targets. No market demand, retention improvement or OCR accuracy has yet been established by this reconstruction.

9 Conclusion

DailyKhata has a focused personal and household expense-management baseline. Its receipt and cash workflows connect to structured accounting records, item-level price history and a data-grounded AI Financial Assistant. D2 inherits the same personas, scope identifiers and priorities, and expands the evidence and elicitation rationale. Research verification and supervisor review remain distinct from document consistency.

10 References

- [S1] DailyKhata. Target Market & Audience Analysis. Original submitted D1, dailykhata-d1.docx and corresponding six-page PDF, 01_Deliverable_1. Undated.
- [S2] DailyKhata. Original submitted D2, dailykhata-d2.pdf, 02_Deliverable_2, 12 pages. Cover repeats the D1 title; body contains problem identification, stakeholder identification and requirement elicitation. Undated.
- [A1] Deliverable 1 task.txt, 01_Deliverable_1/task. User, problem, expectation and target-market assessment criteria. Undated.
- [A2] Deliverable 2 task.txt, 02_Deliverable_2/task. Official D2 title and evidence-based assignment requirements. Stated deadline: 25 August 2026, not proof of submission.
- [G1] DailyKhata Scope Baseline v1.0 and DailyKhata Documentation Standard v1.0, 00_Project_Governance. Both dated 23 September 2026.
- [U1] Project owner instructions for reconstruction, 23 September 2026. Authoritative scope, team identifiers, priorities and constraints.